

Exercises on Chapter 1

Exercise 1 (§1)

Give conditions that a simple graph must satisfy in order that its dual is also a simple graph.

Exercise 2 (§1)

Define an "interval hypergraph" to be a hypergraph whose vertices can be represented by points on a line in such a way that the edges are intervals of the line. Show that if an interval hypergraph is simple then its dual is also an interval hypergraph. Show that a subhypergraph of an interval hypergraph is an interval hypergraph.

Exercise 3 (§1)

For two integers $n \geq r \geq 2$ the r -uniform complete hypergraph of order n is the hypergraph K_n^r whose vertex set is a set X of cardinality n , and whose edges are all the r -subsets of X . What is the rank of K_n^r and of its dual $(K_n^r)^*$?

Exercise 4 (§3)

Let H be an intersecting family of order n , of rank $r = \max_i |E_i|$ and anti-rank $s = \min_i |E_i|$. Hilton [1975] showed that

$$m(H) \leq \sum_{i=s}^r \binom{n-1}{i-1}.$$

Show that this result generalises the Erdős, Chao-Ko, Rado Theorem.

Exercise 5 (§3)

Show how Theorem 6 implies relation (1) of Sperner's Theorem (Theorem 2).

Exercise 6 (§3)

Let $H = \{E_1, E_2, \dots, E_m\}$ be a hypergraph satisfying

$$E_j \not\subseteq E_k \quad (j \neq k)$$

$$E_j \cap E_k \neq \emptyset$$

$$E_j \cup E_k \neq X.$$

Show that $H' = (E_1, E_2, \dots, E_m, X - E_1, \dots, X - E_m)$ is a simple hypergraph. Deduce the following inequality (Schönheim [1968]):

$$m(H) \leq \frac{1}{2} \binom{n}{\lfloor n/2 \rfloor},$$

and this bound is best possible.

Exercise 7 (§3)

Show as in the lemma:

Let $\mathcal{A} = (A_1, A_2, \dots, A_m)$ be a family of m circular intervals on a circle of n points with

- (i) $|A_i| > n/2$;
- (ii) $A_i \cap A_j \neq \emptyset \quad (i \neq j)$
- (iii) $A_i \not\subset A_j \quad (i \neq j)$

Then we have $m \leq n$, with equality if $\mathcal{A}$ is the family $\mathcal{A}_k$ of distinct circular intervals having fixed cardinality $k > \frac{n}{2}$.

Exercise 8 (§3)

Let $\mathcal{A}$ be a family of circular intervals satisfying conditions (2) and (3) of Exercise 7, and for $A \in \mathcal{A}$, put:

$$\begin{aligned} p(A) &= \frac{n - |A| + 1}{|A|} \quad \text{if } |A| \leq \frac{n}{2} \\ &= 1 \quad \text{if } |A| > \frac{n}{2}. \end{aligned}$$

Show that $\sum p(A) \leq n$.

Exercise 9 (§3) (Open Problem)

Let H be a hypergraph on X of order n , let $k \geq 2$ and $t \leq n$ be integers. Erdős and Frankl [1979] conjectured that

$$I \subset \{1, 2, \dots, m\}, \quad |I| = k$$

implies

$$\left| \bigcup_{i \in I} E_i \right| \leq n - t$$

and if m is the maximum with this condition then $H = \{F/F \subset X, |F \cap Y| \leq s\}$ for an integer s and for a set Y of cardinality $t + ks$.

Katona showed that the conjecture is true if $k = 2$, $t \neq 1$. Frankl [1979] showed that the conjecture is true for $k > 2$, $t < \frac{k2^k}{150}$.

Exercise 10 (§4)

Show, using the methods of proof of Theorem 7, that if H is an hereditary hypergraph, the graph $\overline{L(H)}$ (complement of the representative graph) admits a matching covering every vertex, except at most one in each connected component of odd order (Berge [1976]).

Exercise 11 (§5)

Show, using Theorem 5, that the dual of an interval hypergraph has the Helly property.

Exercise 12 (§5)

Consider integers $a_1 < m_1, a_2 < m_2, \dots, a_k < m_k$. Show that the system

$$x \equiv a_i \pmod{m_i} \text{ for } i = 1, 2, \dots, k$$

has a solution x if and only if every pair (i, j) with $1 \leq i < j \leq k$ satisfies

$$a_i \equiv a_j \pmod{\text{lcm}(m_i, m_j)}.$$

(Use Corollary to Theorem 10).

Exercise 13 (§5)

Show that for $k \geq 3$, every simple graph is k -Helly.

Exercise 14 (§6)

Using Frankl's lemma (Lemma 2 of Theorem 14), prove the following result, due to Lovász (which generalises the corollary to Theorem 14):

Let H be an r -uniform hypergraph and let x be a positive real number such that

$$m(H) = \frac{x(x-1)\dots(x-r+1)}{r!}$$

Then

$$m([H]_{r-1}) \geq \frac{x(x-1)\dots(x-r+2)}{(r-1)!}.$$

Exercise 15 (§8)

Let $d(m)$ be the minimum cardinality of a set X having the property that every graph of order m is the representative graph of at least m distinct subsets of X . Show (by induction on m) that

$$d(2) = 2$$

$$d(3) = 3$$

$$d(m) = \lceil \frac{m^2}{4} \rceil \text{ if } m \geq 4$$

(Erdős, Goodman, Pósa [1966]).